LANCASHIRE TEACHING HOSPITALS NHS FOUNDATION TRUST · RESULTS REPORT-OUT

Which trials pharmacy can complete within 60 days, and what limits that number

THE FINDING

On-time performance is set by the usable time remaining when the study reaches pharmacy — not by the number of set-ups opened each month.

Optimisation identifies the best portfolio within the time available. Simulation tests whether that promise survives the variation in set-up durations. Sensitivity analysis identifies which operational changes genuinely move the outcome.

The clock opens before pharmacy is notified, and that gap consumes the target

60

days in the target window, from the later of HRA approval and site selection

79

days is the median pharmacy work interval — longer than the whole target before any hand-over is counted

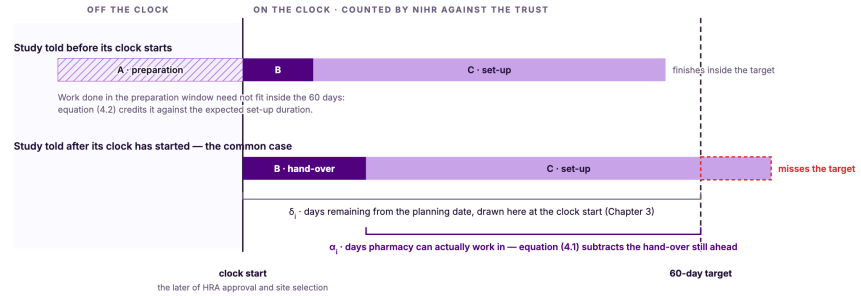
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case studies finished inside the target from a standing start

FIGURE 4.2 · THE THREE SEGMENTS OF THE SET-UP WINDOW

Hand-over consumes target days; preparation costs the target nothing

Two schematic studies. The clock starts at the later of HRA approval and site selection; time to the left of it is not counted by NIHR, time to the right of it is. Bar lengths are illustrative, not measured.



Schematic of the model's structure, not a measurement. Segment definitions: Table 4.3. Demonstration-set timings: Appendix 4.D.

Figure 4.2 — where the 60 days are spent

The four trials that met the target had all entered the pharmacy queue **before the clock opened**. The window did not stretch; the work began earlier.

The binding constraint is the usable time remaining when the study reaches pharmacy — and that is decided before any capacity decision is made.

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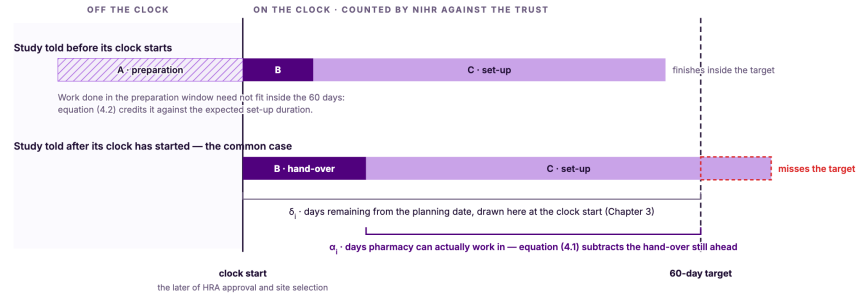
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Three sources from the trust, and what each can and cannot supply

SOURCE	WHAT IT SUPPLIES	ROLE AND LIMITATION
CTIMP 150-day tracker	Problem scale and arrival pattern	104 trials, of which 23 carry a measurable pharmacy duration
Prioritisation Tool, version 1.6	Measure 4b, operational burden for pharmacy and aseptics, re-scored jointly by both teams. It is the pharmacy burden rank used throughout	Its value score is not used, so no trial is selected or excluded on perceived value
The August 2026 case-study return	Every per-study planning parameter: the dates, the pathway, the burden rank	20 completed CTIMPs, 19 usable for the planning replay

WHAT THE EVIDENCE RESTS ON, SAID ONCE

No system captures the pharmacy set-up start automatically. Both teams reconstructed each date from their own records for the return, which is why the dissertation treats them as retrospective estimates. Hands-on approval time is recorded nowhere at all, so it is carried as a sensitivity parameter rather than a measured value.

The set-up duration floors are demonstrated-achievable standards rather than averages or forecasts. Results computed on them describe an **optimistic operating condition, not an expected one**. The measurement stage of the roadmap is how the trust closes both limitations.

Optimisation finds the best plan, simulation tests whether it holds, sensitivity analysis finds what matters

Monthly trust queue → **Optimisation** → Best on-time portfolio → **Simulation** → **Reliable commitment**

Optimisation

“If we must decide now, which waiting trials should we commit to, so as to complete the most inside 60 days?” Where plans tie, it prefers the one leaving the least weighted lateness across the queue, so deferring a trial carries an explicit cost.

Simulation

“If set-up duration varies as it has historically, how often will the selected plan actually deliver what it promised?” Each committed plan was replayed 10,000 times.

Sensitivity analysis

“Which operational changes genuinely change the outcome?” Six settings were swept across their tested ranges at all 25 decision points.

The model decides which trials to commit. It does not decide the order in which the work is done, and it does not rank a study's worth. All eligible trials are treated as equally worthwhile.

If nothing changes: four trials were ever feasible, and all four entered the queue before the clock opened

9 of 25

monthly decision points already admit one on-time trial, with no change at all

4

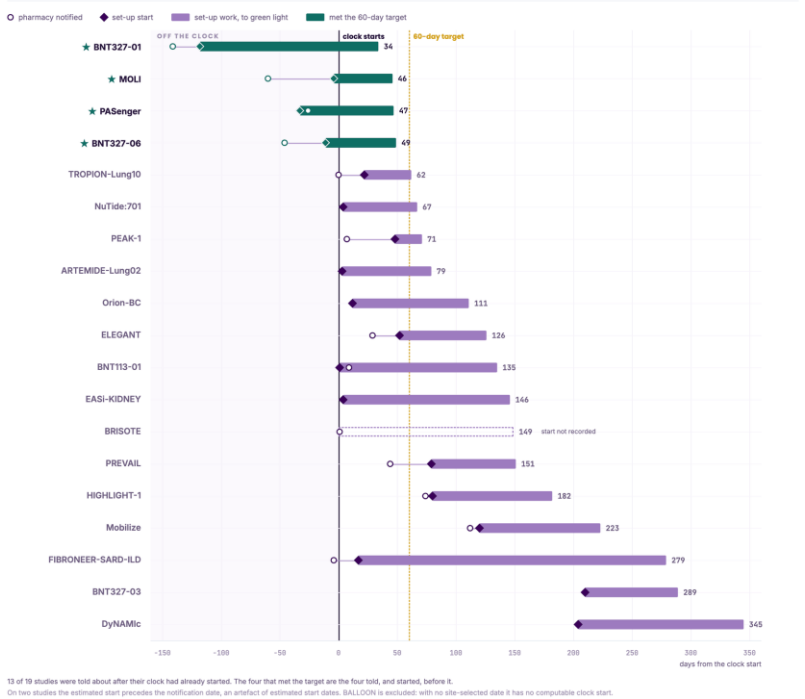
distinct trials in total — a trial that still fits is re-committed and counted again

The model identifies those four **in advance rather than in hindsight**: they are exactly the four that met the target in reality. It does not uncover any additional trial the service could have brought in on time.

TRIAL	DAYS ENTERED BEFORE CLOCK START	WORK INTERVAL (DAYS)	CLOCK INTERVAL (DAYS)
MOLI	4	50	46
BNT327-06	11	60	49
PASenger	33	80	47
BNT327-01	118	152	34

Two of the four met the target despite requiring more than 60 days of work, because their set-up began before the window opened. Where the plan falls short it does so by relatively little: four trials missed by 20 days or less and, under the plan, fall 10 to 29 days short of their available windows. No selection rule recovers those days.

Where the 60 days go: every case study on one clock



The four-trial figure is an **upper bound** across the whole period under the recorded timing conditions**. It is not a count per cycle, and not a throughput estimate.**

More slots and better selection reduce lateness, but they do not create time

LEVER	RAISES THE CEILING?	WHAT IT CHANGES	MANAGEMENT IMPLICATION
Preparation before the clock opens	**Yes**	Creates usable time	**The primary intervention**
Pharmacy queue, notification to start	Partly	Rescues borderline moderate cases	Prioritise after notification
Hand-over gap, clock to notification	Limited	Reduces lateness most, by 498 days	Improve jointly, but do not rely on it alone
Set-ups opened per month	**No**	Reduces lateness only	Do not justify this as an on-time intervention
Optimised selection	Limited	Helps only under very tight capacity	Decision support, not the root-cause solution
Hands-on days per approval	Uncertain	Can become binding if slow	Measure it before operationalising the planner

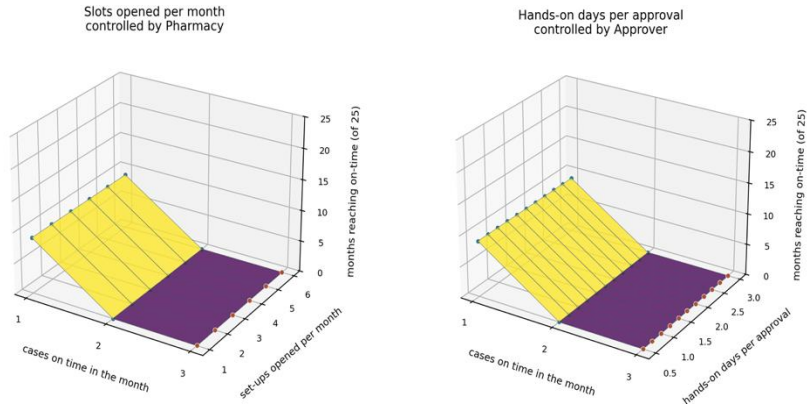


Figure G.2 — flat across the whole range

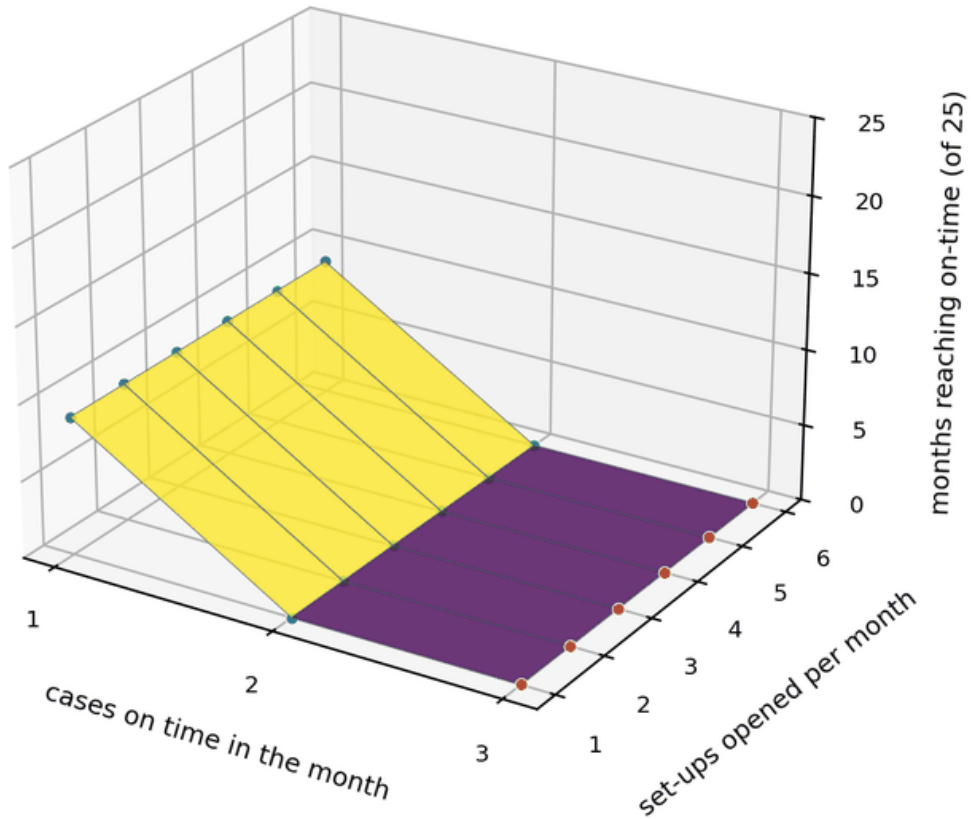
The approval signature isn't binding under the assumed workload, and we don't measure that input anywhere. At 1.5 hands-on days, the expected wait reaches 48.5 days with six commitments; at three days, 97 days with only three. Which is why recording it sits on the roadmap.

9 → 17
months with at least one on-time trial, of 25, when a 30-day preparation lead is given

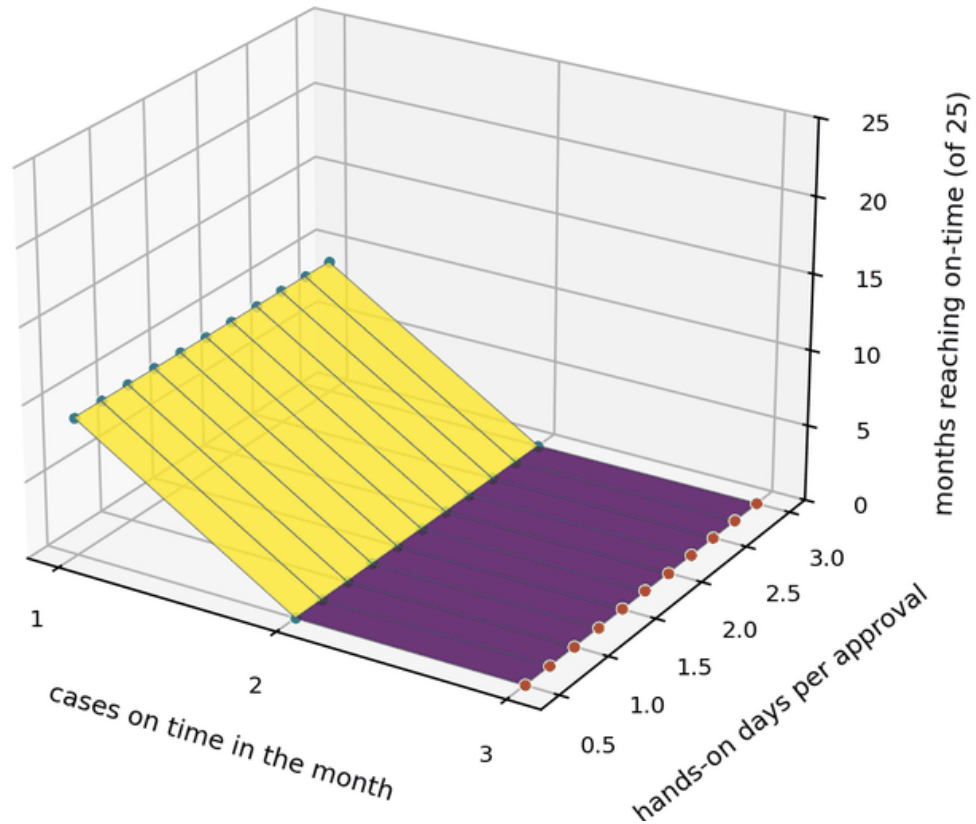
+5
further decision points gain an on-time completion when the pharmacy queue is shortened

0 of 25
decision points gain an on-time completion from opening more set-ups

Slots opened per month
controlled by Pharmacy



Hands-on days per approval
controlled by Approver



Preparation is the only tested lever that creates time rather than redistributing it

11
months of 25 reach one on-time trial at a 10-day lead

17
at a 30-day lead — against a baseline of 9

20
at a 60-day lead

— Two on-time trials are harder: no month reaches two below 10 days of lead, seven reach it at 30 days and twelve at 60. Three occur in one month at 30 days and two at 60.

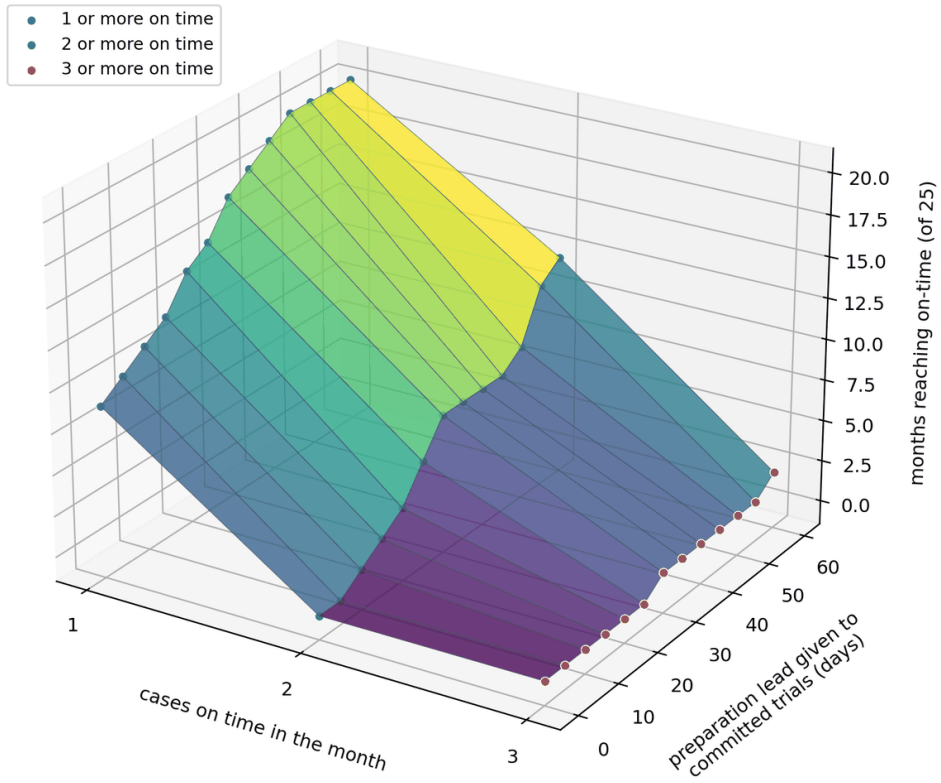
— Every other setting redistributes days inside the window. Preparation adds days **before it opens**, which is why it is the only one that moves the ceiling.

— The first rescue is cheap; each further one costs more lead than the last.

Caveat. The base model does not charge pre-window preparation against the commitment limit, so it may overstate the benefit. Tested under reduced capacity, preparation still increases the number of months with at least one on-time completion, although the three-trial result disappears with two setups.

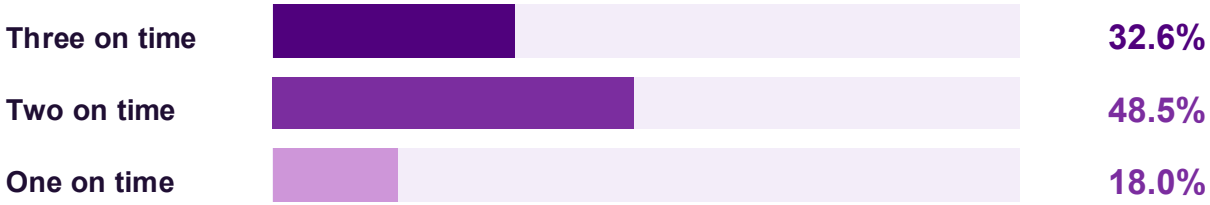
The honest limit. What a queue can reach is set by its own composition. Across the whole range tested, the February 2025 queue never passes two. That ceiling belongs to the queue rather than to any lever.

How far preparation carries the portfolio
all 25 monthly queues, standard times



Three on-time trials is a feasible maximum, not a reliable service commitment

Each committed plan was replayed **10,000 times** against the observed spread of set-up durations. On the synthetic standing queue, the plan that promises three delivers:



Committing to 3 studies simultaneously, the test indicates that **at least a 90%** chance of delivering **at least 1 study** on time

> **Agree a confidence threshold, and commit to the count the service can hold at it — not to the maximum the deterministic model reports.**

— Where a margin is smaller than the observed spread of durations, the result is a tie-break rather than a finding. A one-day lateness label carries no information.

— Every figure here understates total uncertainty, because the approval wait is held at its expected value throughout.

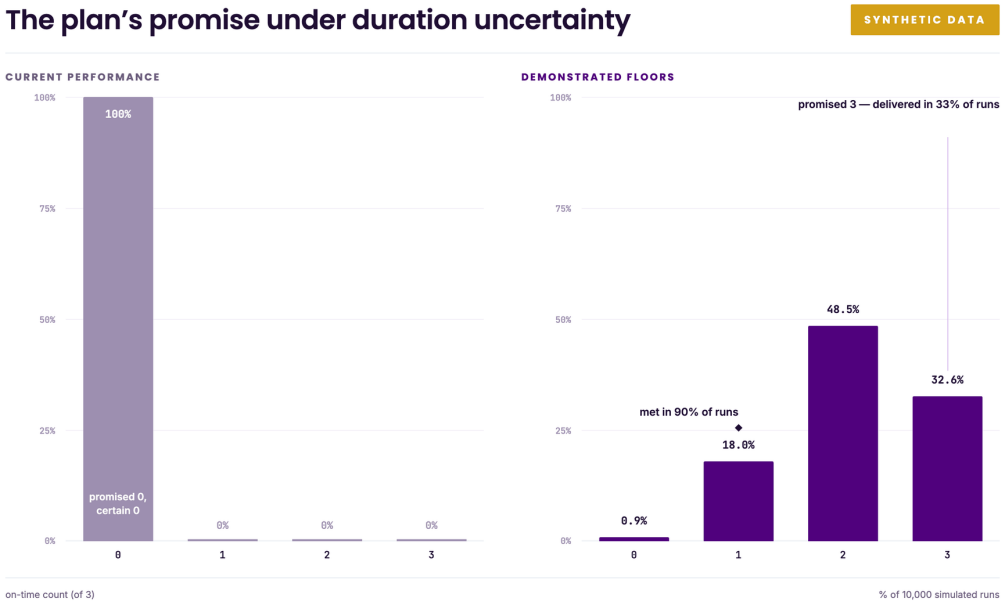
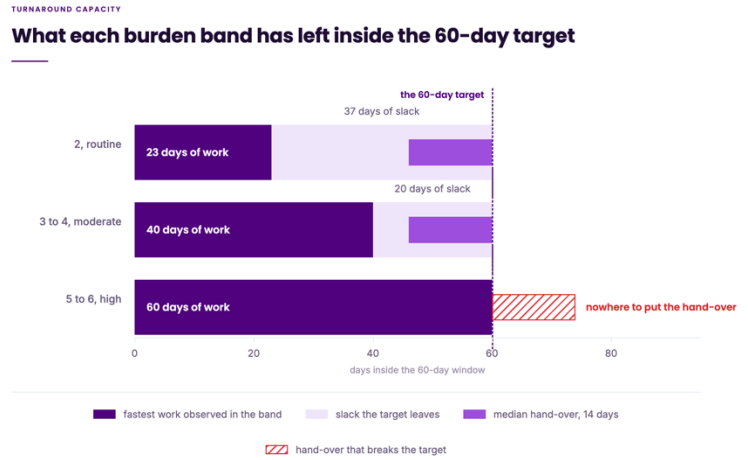


Figure 5.3 — the plan's promise under duration uncertainty

Preparation does not have to be given to every trial, and it should not be

The slack each band carries inside the 60-day window, from the fastest set-up each has achieved:

BAND	FASTEST OBSERVED	SLACK IN THE WINDOW	ACTION
Low, rank 2	23 days	37 days	Normal path — needs neither intervention
Moderate, ranks 3 to 4	40 days	20 days	Front of the pharmacy queue after notification
High, ranks 5 to 6	60 days	**none**	**Preparation lead before the clock opens**



A high-burden study **cannot be rescued by any choice made after the clock starts**. Against a median 14-day hand-over it arrives with no room, so preparation is not an optimisation for these studies — it is the only route by which they can be on time. A moderate study is on time if the hand-over is quick and lost if it is not; that is the band where the pharmacy queue does the work. A low-burden study carries 37 days and needs neither.

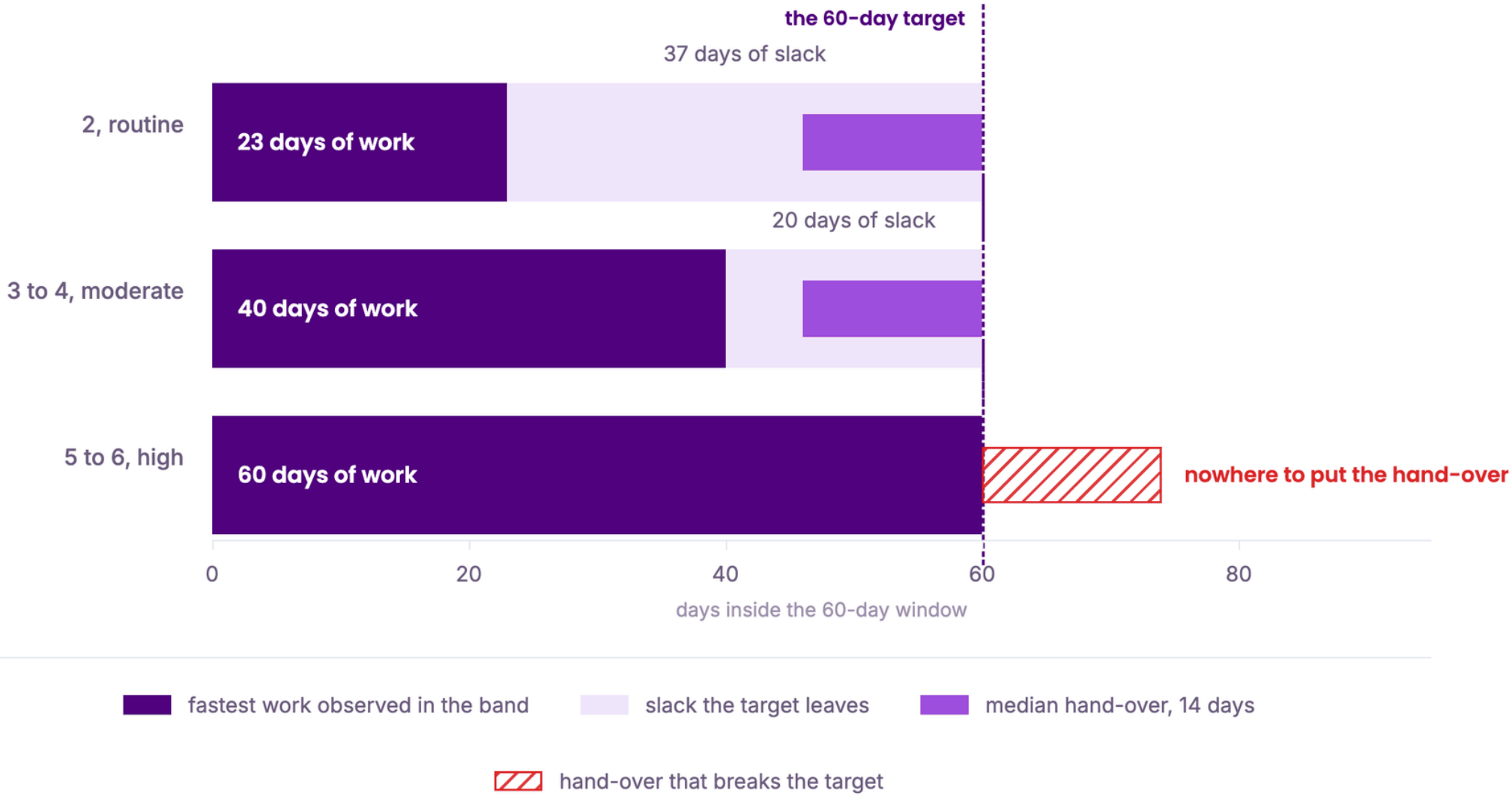
Say it as a floor, not an average. What the data supports is not that high-burden studies take longer on average, but that **none has ever been set up here in under 60 days**. The identifier is available at notification: all six high-burden studies in the case set are aseptic, so the aseptic requirement catches every one of them, plus five others. It is a risk screen, not a value ranking, and it does not decide which studies are committed.

LEAD BEFORE THE CLOCK OPENS	FINISHES ON TIME
None	50 per cent
28 days	**75 per cent**
63 days	90 per cent

Twenty-eight days is the working standard. Sixty-three is a second window on top of the first, and is worth it only for a study that must land.

TURNAROUND CAPACITY

What each burden band has left inside the 60-day target



Score the field the trust already owns, and the rule runs itself

Measure 4b of the Prioritisation Tool is the pharmacy burden score. It reads **0 on every study scored to date** — its lowest-burden option.

Zero on every study is not a finding about the portfolio. The scoring group has no pharmacy member, so when pharmacy is involved, nobody in the room can judge the burden. **The measure is uncollected rather than flat.** That is why the study used a burden rank returned separately by pharmacy, and it is the one change that would let the trust run the triage rule itself.

TWO CHANGES, NEITHER NEEDING NEW DATA COLLECTION

Put a pharmacy clinical trials voice in the scoring group, for measure 4b only. It is the single measure drawing on pharmacy clinical trials, technical services, quality assurance, clinical and supply, and ePMA together, so nobody else in the room can score it. It costs a standing invitation to one meeting.

Adopt the rubric on the leave-behind, which maps what a study asks of pharmacy onto a rank, and the rank onto the preparation it needs. Once 4b is scored at the point the tool is used, the rule runs automatically — the tool already fires before set-up begins, which is exactly when the lead has to be granted.

TARGETING IS WHAT MAKES IT AFFORDABLE

APPROACH	MONTHS ON TIME, OF 25	Preparation (days)
No lead, today	9	0
Targeted, 35-day lead on high-burden studies only	**14**	**735**
Spread, 21-day lead across the queue	14	1,407
Targeted, 60-day lead on high-burden studies only	16	1,260

The same 14 months, for a little over half the R&I effort. Across the whole range, targeting costs about 147 R&I days for each additional month that delivers, against 234 to 365 days when the lead is spread.

Say the limit plainly. Spreading the lead is the only way to reach three on-time trials in a month; targeting reaches two. Targeting is the affordable policy, not the maximum one. It is recommended because it is the version the service can actually run.

On the ranks. The burden ranks were given after the outcomes were known. They classify what these twenty studies demanded rather than predict what a new one will, so the rubric is a risk screen to revise as pharmacy scores real studies against it.

Start targeted preparation now, make the score operational next, then measure the process

NOW · 0 TO 3 MONTHS
Create time for studies with no slack

Give high-burden and aseptic-flagged studies a 28-day preparation lead before the clock opens. The only tested lever that raises the ceiling, and for these studies the only route to on time at all.

Move moderate-burden studies to the front of the pharmacy queue after notification. That segment never exceeds 23 days and sits inside pharmacy's own control.

Commit to a count the service can hold, at a stated confidence. It changes what is promised, not what is done.

NEXT · 3 TO 6 MONTHS
Make the rule run without an analyst

Add a pharmacy clinical trials representative to the 4b scoring group. It costs a standing invitation to one meeting.

Score studies before set-up begins, using the rubric, rather than classifying them retrospectively.

Link the score to the action rule: high burden takes the lead, moderate takes queue priority, low takes the normal path. The tool already fires before set-up begins, which is when the lead has to be granted.

THEN · 6 TO 12 MONTHS
Replace estimates with records

Record four timestamps at the time: pharmacy notified, set-up started, approval-ready, green light issued. Durations become records rather than reconstructions.

Re-estimate durations, validate the rubric prospectively, and re-run the simulation on observed data. The approval-ready timestamp is the one input measured nowhere: the assumption holds below 1.5 hands-on days and breaks sharply above it.

Build the planner only once data quality is sufficient — before that it would compute on reconstructed dates.

NOT RECOMMENDED — AND NOT A FOURTH STAGE

× Opening more set-ups per month to chase the target. It changed the on-time count at none of the 25 decision points.
× Closing the hand-over gap entirely. It runs to 210 days and is not wholly inside anyone's control.
× Changing the selection rule on its own. From three set-ups upwards it selects the same trials as arrival order.
× Building the planner first. It belongs after the measurement stage, not before it.

Staffing is held constant throughout the analysis, so every gain above comes from the establishment the establishment already has. What it is worth in money the study cannot say, because commercial income per trial and the cost of a set-up day are not in the data. The table converts directly once the trust puts those two figures against it.

Four sentences to take away

01 **The problem is not how many studies pharmacy opens each month. It is the usable time remaining when the study reaches pharmacy.**

02 **Optimisation tells us the best possible portfolio within that time. Simulation tells us what the service can responsibly promise.**

03 **More capacity and better selection reduce lateness, but only earlier preparation raises the on-time ceiling.**

04 **Start with targeted preparation for high-burden studies, operationalise the burden score, then measure the timestamps that make the rule precise.**

Do not add slots to chase on-time performance. Create earlier preparation for the studies that arrive with no slack, then measure the process properly.

These results describe **demonstrated-achievable conditions**, not average expected performance, on twenty case studies with reconstructed set-up dates. The next phase is to replace those estimates with recorded operational timestamps. **The one-page rule follows as a leave-behind.**

PHARMACY CLINICAL TRIALS SET-UP · LTHTR

The rule, small enough to put on the wall

Step 1. At notification, ask three questions

All three are answerable from what R&I already holds.

1 Does the study need aseptic preparation?

2 What is the later of HRA approval and site selection? That day is when the 60-day clock opens.

3 Has pharmacy been told before that day?

Step 4. When committing the month, state a count the service can hold

A plan promising three on-time completions delivers three in about a third of runs and two in about half. Commit to the count that holds at the confidence the service has agreed, not to the maximum the plan reports.

Step 5. Review it at the meeting that already happens

The screen needs the study name, the two approval dates, the pathway and the band. It supports judgement and does not replace governance.

Step 2. Score measure 4b using this rubric

RANK	WHAT THE STUDY ASKS OF PHARMACY	BAND
2	Dispensary-led, low-complexity IMP, no aseptic work, no new prescribing build	Low
3 to 4	Either aseptic preparation or an electronic prescribing build, but not both; standard supply arrangements	Moderate
5 to 6	Aseptic preparation with a prescribing build, non-standard supply, or a technical services or quality assurance requirement the unit has not run before	High

Note for the room: this rubric is the study's ranks 2 to 6 written as descriptions. The ranks were given after the outcomes were known, so they classify what these twenty studies demanded rather than predict what a new one will. Use it as a risk screen and revise the wording as pharmacy scores real studies against it.

Step 3. Act on the band

High, 5 to 6

Begin pharmacy preparation as soon as the study is known, targeting 28 days before the clock opens. Do not wait for the hand-over meeting.

No slack inside the window. Without a lead the study is on time half the time; with 28 days, three times in four.

Moderate, 3 to 4

Normal path, but move it to the front of the pharmacy queue after notification.

Twenty days of slack against a median 14-day hand-over. The queue is what decides it.

Low, 2

Normal path.

Thirty-seven days of slack. Neither intervention changes the outcome.

Back-up slides

Seven items held in reserve: the model parameters and duration floors, the full lever table, the approval surface, the statistical tests, what a rescue costs, the robustness sweep and burden weights, and the planner interface.

The model parameters, and the pathway duration floors

INPUT	VALUE USED	WHERE IT COMES FROM
Target window	60 days from the later of HRA approval and site selection	The national 150-day definition
Time available	Due date less the later of today and the date pharmacy can begin	Calculated per trial; delay still ahead is subtracted, delay already served is not
Set-up duration floors	30 days prescription-only, 50 routine, 60 both, 72 aseptic-only, with a 63-day overlay where a trial is both high burden and on the Both pathway	The demonstrated work floors, audited in Appendix F
Preparation lead	Tested at 0, 7, 14, 21, 35 and 60 days; zero in the base case	Tested as a range, not assumed
Approval wait	Rises with the number of trials committed at once	A single-server queue formula. The only input in the results that is not measured
Set-ups per cycle	Tested at 1 to 6, against a stated limit of two	The service-level agreement

The three hand-over medians, and why they differ. The CTIMP tracker gives a median 8-day hand-over across 104 trials. The case-study return gives a median of 35 days before pharmacy began, on the 14 of 19 usable studies where the hand-over was positive, running as high as 210. The triage rule uses the median 14 days between the clock starting and pharmacy receiving the study. Each is valid for its own source and definition; only the 14-day figure is used in the rule.

Table G.2 in full: each setting's tested range and what it moves

SETTING	TESTED RANGE	MONTHS ON TIME, OF 25	BEST COUNT	LARGEST SINGLE-DATE REDUCTION IN WEIGHTED LATENESS
Preparation lead on committed trials	0, 7, 14, 21, 35 and 60 days	**20**	**3**	The only setting that raises the on-time ceiling
Pharmacy queue, notification to start	Never exceeds 23 days in the records	14	2	About one-seventh of the hand-over gap's reduction, but adds on-time completions at five further decision points
Hand-over gap, clock to notification	Swept to 120 days; recorded gaps run to 210	11	2	498 days, the largest of any setting, but only two further on-time decision points
Target window	55 to 70 days	11	2	Held by national policy, not by the trust
Set-ups opened per month	1 to 6, against a stated limit of two	9	1	Lowers weighted lateness at every on-time count, changes the on-time count at none
Hands-on days per approval	0.5 to 3 days	9	1	Flat across the range at the assumed workload; measured nowhere

The baseline is nine months. On-time completion is a threshold measure while lateness accumulates, which is why the two outcome columns rank the settings differently.

What if the signature takes longer than a day

— Below 1.5 hands-on days all six set-ups are worth opening.

— At two days about four are worth opening.

— At three days only two are worth opening.

— This is where the approval assumption breaks, which is why it sits in back-up, and why the approval-ready timestamp is in the measurement stage of the roadmap.

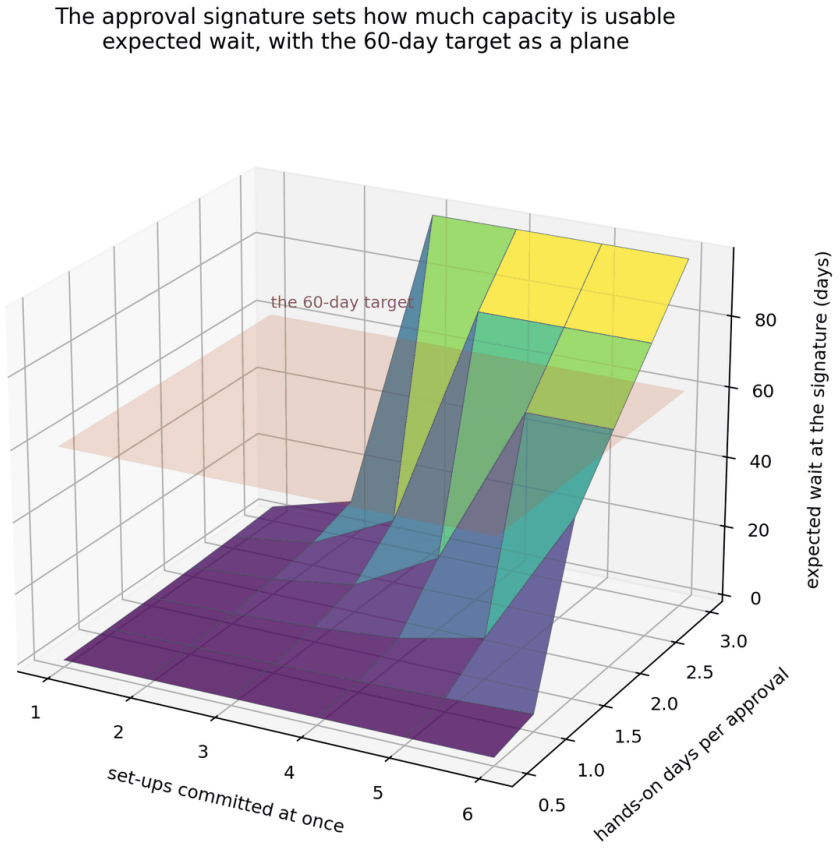


Figure G.3 — the approval surface

The statistical tests, and what each does and does not establish

p = 0.557

Kruskal–Wallis, across the three burden bands

The bands showed no detectable difference in median duration, although groups this small would not reveal a moderate one. What separates them is the fastest each has achieved, which is what sets the slack.

p = 0.067

Fisher's exact, on the six high-burden trials

The two that began before the clock started met the target; all four that began afterwards missed it. The pattern fits the proposed mechanism but does not establish it.

p = 0.71

Work intervals, early against late starters

Work intervals did not differ significantly. Earlier preparation moves work outside the window rather than shortening it. Given the small case base, this does not establish that earlier preparation has no operational effect.

Two further observations from the case set. No recorded trial characteristic predicted an early start, so the early starts were not the product of a rule the service could repeat. The model also starts the right trials sooner: three of the four are among the six it starts earliest, by 34, 22 and 17 days, and of the twelve trials the service began before the model would, eleven missed the target anyway.

What a rescue costs, what preparation buys, and what capacity buys

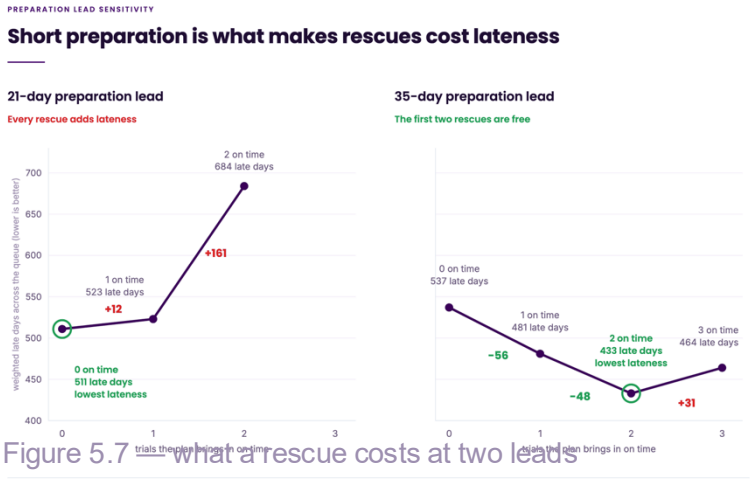


Figure 5.7 — what a rescue costs at two leads

1 May 2025 queue, seven trials, six set-ups, demonstrated floors with the high-burden overlay. Best plan at each on-time count.
At 21 days no plan reaches three on time. On the 1 February 2025 queue the two objectives do not conflict at any lead.



Figure 5.8 — what preparation buys

Least preparation lead that first reaches each on-time count, six set-ups available, demonstrated floors with the high-burden overlay.
At every step the plan commits exactly as many set-ups as trials it rescues. Committed burden is Figure 5.10.

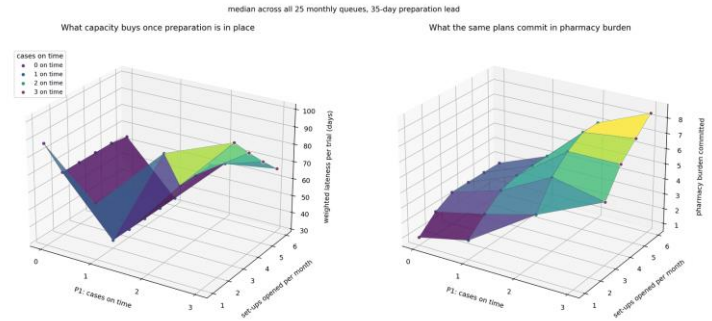


Figure 5.9 — what capacity buys

— On the largest queue, 1 May 2025, a 21-day lead makes each rescue cost additional lateness. A 35-day lead brings two trials in on time at the lowest weighted lateness of any plan. On the 1 February 2025 queue the two goals never conflict at any lead. A governance rule that ranks lateness first therefore gives up nothing when preparation is adequate and two rescues when it is not.

— Figure 5.8 reports the least preparation that first reaches each on-time count. The first rescue is cheap; each further one costs more lead than the last. Across the whole range tested the February queue never passes two, and that ceiling belongs to the queue rather than to any lever.

— Holding the lead at 35 days, weighted lateness falls with every slot opened at each on-time count, while the pharmacy burden the plan commits rises with both. That is the capacity decision stated properly: it buys a shorter queue, not a higher on-time count.

The robustness sweep, and the burden weights

9,000 GENERATED PORTFOLIOS

— At current performance the model never beats the arrival-order rule for on-time completions.

— At the demonstrated floors it wins in 36 per cent of cases, each win costing 24.3 weighted days of lateness elsewhere.

— It helps most when trials arrive early and the aseptic share is low.

ROUTINE-TRIAL-EQUIVALENT WEIGHTS

1.00	1.74	2.61
Low burden	Moderate burden	High burden

Used only to compare the burden composition of two portfolios.

They are not staff-time ratios.

The planner interface, with the caveat that it computes nothing and has not been used by anyone

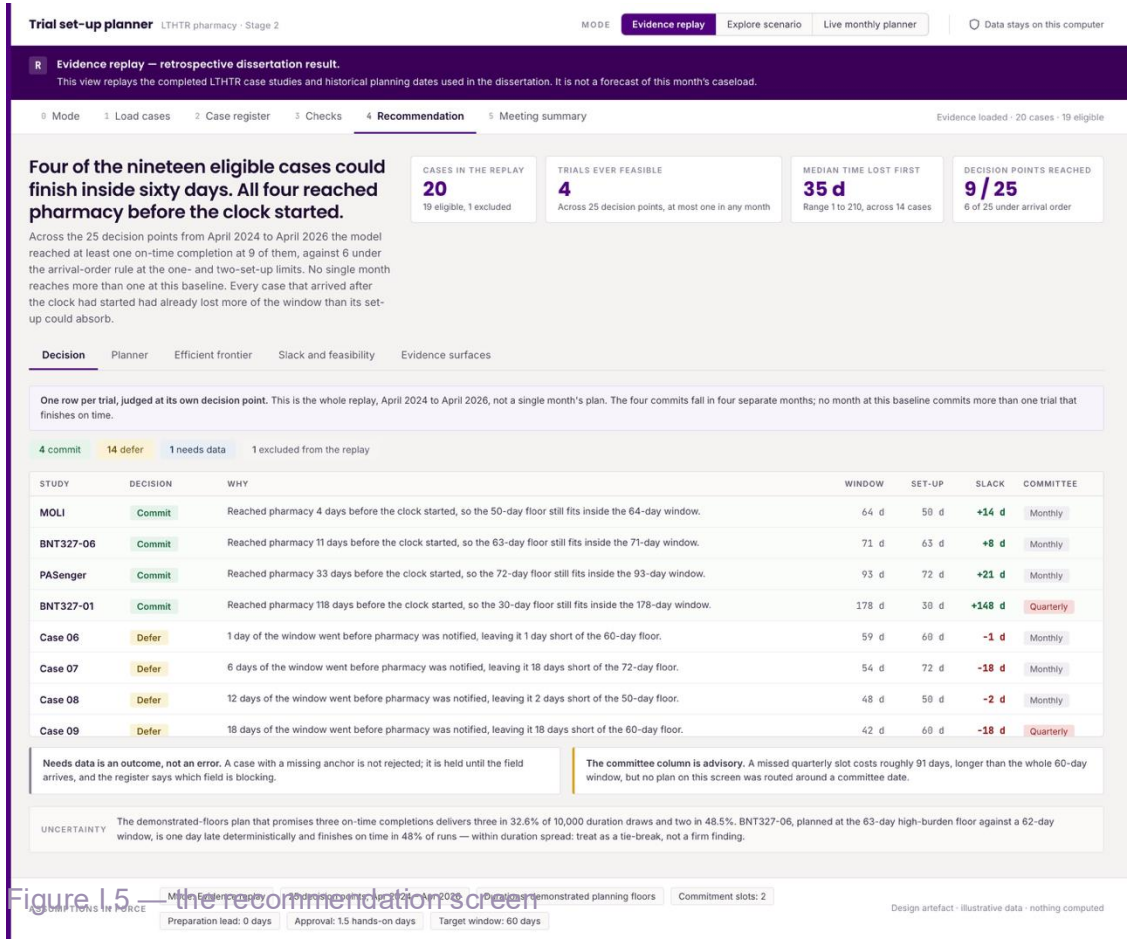


Figure I.5 — the recommendation screen

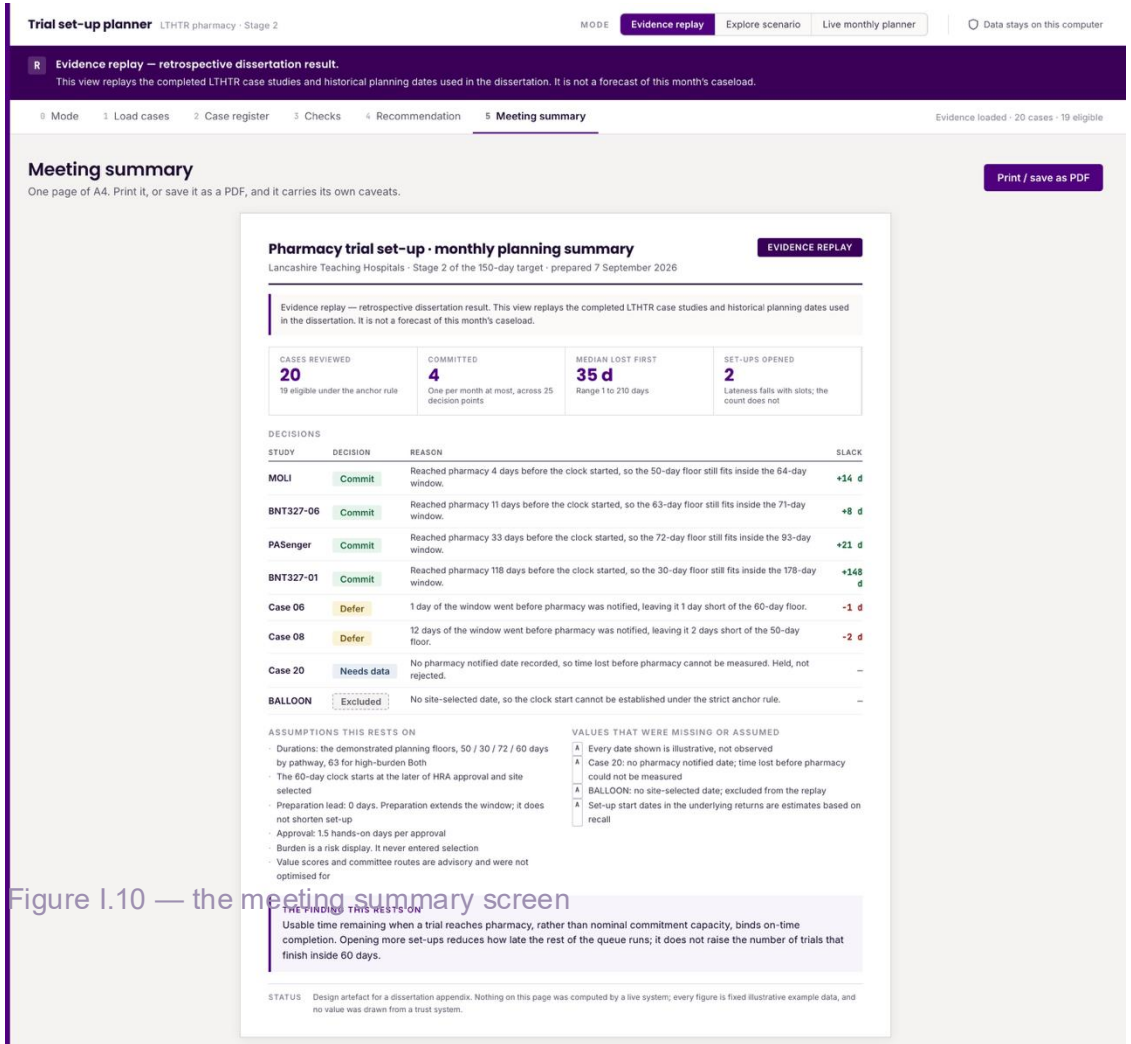


Figure I.10 — the meeting summary screen